Lab7 Fuzz Testing

highway_toll.py

highway_toll.py > parse_plate

```python
def parse_plate(plate):
    state, district, series, number = plate.split('-')
    return state.upper(), int(district), int(number)

def trip_cost(text):
    count, rate = text.lower().split('x')
    return int(count) * int(rate)

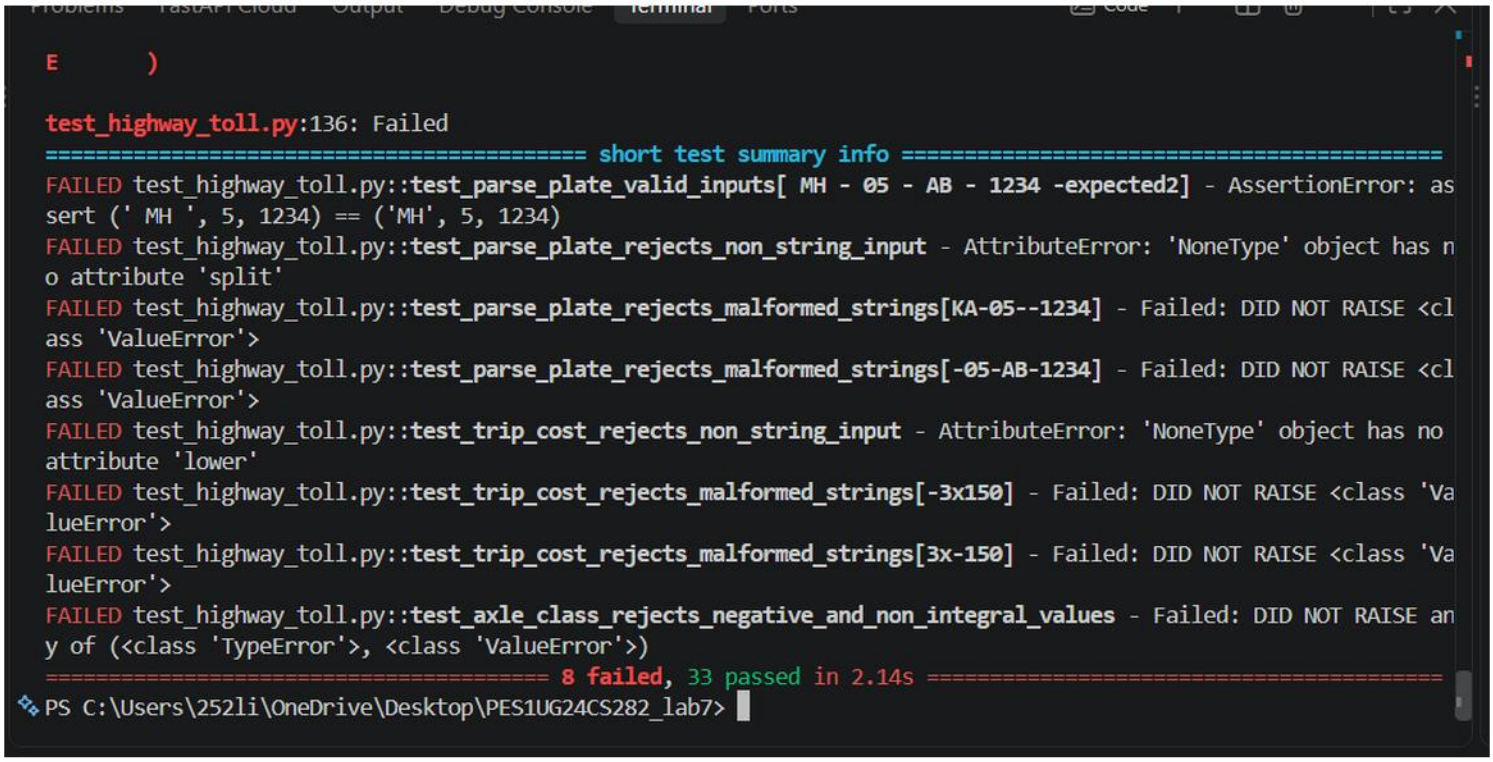
def axle_class(axles):
    n = int(axles)
    if n == 2:
        return 'car'
    if n <= 4:
```

Problems FastAPI Cloud Output Debug Console Terminal Ports Code

```
PS C:\Users\252li\OneDrive\Desktop\PES1UG24CS282_lab7> python -u "c:\Users\252li\OneDrive\Desktop\PES1UG24CS282_lab7\highway_toll.py"
==== Running parse_plate ====
Enter a plate (e.g. KA-05-AB-1234): KA-05-AB-1234
State, District and Number: ('KA', 5, 1234)

==== Running trip_cost ====
Enter trips x rate (e.g. 3x150): 3x150
Trip Cost: 450

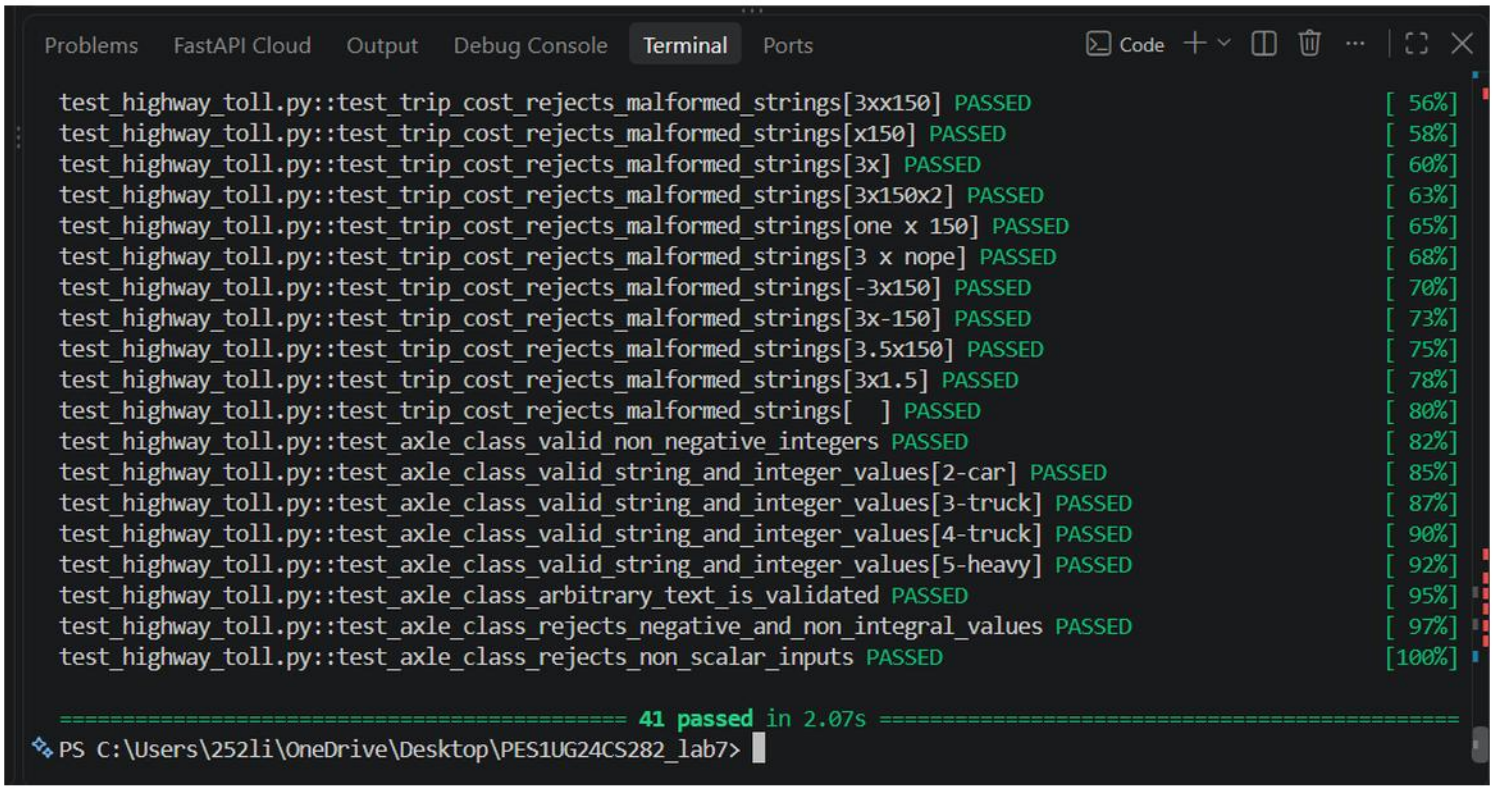
==== Running axle_class ====
Enter an axle count (e.g. 3): 3
Vehicle Class: truck
PS C:\Users\252li\OneDrive\Desktop\PES1UG24CS282_lab7>
```


E)
test_highway_toll.py:136: Failed
short test summary info
FAILED test_highway_toll.py::test_parse_plate_valid_inputs[MH - 05 - AB - 1234 -expected2] - AssertionError: assert (' MH ', 5, 1234) == ('MH', 5, 1234)
FAILED test_highway_toll.py::test_parse_plate_rejects_non_string_input - AttributeError: 'NoneType' object has no attribute 'split'
FAILED test_highway_toll.py::test_parse_plate_rejects_malformed_strings[KA-05--1234] - Failed: DID NOT RAISE <class 'ValueError'>
FAILED test_highway_toll.py::test_parse_plate_rejects_malformed_strings[-05-AB-1234] - Failed: DID NOT RAISE <class 'ValueError'>
FAILED test_highway_toll.py::test_trip_cost_rejects_non_string_input - AttributeError: 'NoneType' object has no attribute 'lower'
FAILED test_highway_toll.py::test_trip_cost_rejects_malformed_strings[-3x150] - Failed: DID NOT RAISE <class 'ValueError'>
FAILED test_highway_toll.py::test_trip_cost_rejects_malformed_strings[3x-150] - Failed: DID NOT RAISE <class 'ValueError'>
FAILED test_highway_toll.py::test_axle_class_rejects_negative_and_non_integral_values - Failed: DID NOT RAISE any of (<class 'TypeError'>, <class 'ValueError'>)
8 failed, 33 passed in 2.14s
PS C:\Users\252li\OneDrive\Desktop\PES1UG24CS282_lab7>

```
========================================= 41 passed in 2.07s =========================================
PS C:\Users\252li\OneDrive\Desktop\PES1UG24CS282_lab7> python highway_toll.py
==== Running parse_plate ====
Enter a plate (e.g. KA-05-AB-1234): KA-05-AB-1234
State, District and Number: ('KA', 5, 1234)

==== Running trip_cost ====
Enter trips x rate (e.g. 3x150): 3x150
Trip Cost: 450

==== Running axle_class ====
Enter an axle count (e.g. 3): 3
Vehicle Class: truck
PS C:\Users\252li\OneDrive\Desktop\PES1UG24CS282_lab7>
```


Problems
FastAPI Cloud
Output
Debug Console
Terminal
Ports
Code
test_highway_toll.py::test_trip_cost_rejects_malformed_strings[3xx150] PASSED [56%]
test_highway_toll.py::test_trip_cost_rejects_malformed_strings[x150] PASSED [58%]
test_highway_toll.py::test_trip_cost_rejects_malformed_strings[3x] PASSED [60%]
test_highway_toll.py::test_trip_cost_rejects_malformed_strings[3x150x2] PASSED [63%]
test_highway_toll.py::test_trip_cost_rejects_malformed_strings[one x 150] PASSED [65%]
test_highway_toll.py::test_trip_cost_rejects_malformed_strings[3 x nope] PASSED [68%]
test_highway_toll.py::test_trip_cost_rejects_malformed_strings[-3x150] PASSED [70%]
test_highway_toll.py::test_trip_cost_rejects_malformed_strings[3x-150] PASSED [73%]
test_highway_toll.py::test_trip_cost_rejects_malformed_strings[3.5x150] PASSED [75%]
test_highway_toll.py::test_trip_cost_rejects_malformed_strings[3x1.5] PASSED [78%]
test_highway_toll.py::test_trip_cost_rejects_malformed_strings[ ] PASSED [80%]
test_highway_toll.py::test_axle_class_valid_non_negative_integers PASSED [82%]
test_highway_toll.py::test_axle_class_valid_string_and_integer_values[2-car] PASSED [85%]
test_highway_toll.py::test_axle_class_valid_string_and_integer_values[3-truck] PASSED [87%]
test_highway_toll.py::test_axle_class_valid_string_and_integer_values[4-truck] PASSED [90%]
test_highway_toll.py::test_axle_class_valid_string_and_integer_values[5-heavy] PASSED [92%]
test_highway_toll.py::test_axle_class_arbitrary_text_is_validated PASSED [95%]
test_highway_toll.py::test_axle_class_rejects_negative_and_non_integral_values PASSED [97%]
test_highway_toll.py::test_axle_class_rejects_non_scalar_inputs PASSED [100%]
41 passed in 2.07s
PS C:\Users\252li\OneDrive\Desktop\PES1UG24CS282_lab7>